

Cross-Platform Autonomous Remediation Blueprint

Target Infrastructure: Multi-OS (Windows, Linux, macOS), Hybrid Active Directory, Multi-Cloud (AWS, Azure, GCP), Multi-SIEM/EDR API Aggregation Layer.

Execution Mode: Fully Autonomous (Zero-Trust Machine-Gate Orchestration).

1. Global Threat Intel & Cross-Platform Orchestration Matrix

Below is the consolidated blueprint mapping active findings to cross-platform decoupled remediation micro-agents operating via native cloud APIs and OS hooks.

Threat / Target	Owner / Progress	Risk	Multi-OS / Hybrid-Cloud Autonomous Remediation Vector
WMI Lateral Movement	Marta Kowalski 72% (Active)	82	Windows AD/WinRM Engine: Dispatches autonomous decoupled loop evaluating root/subscription. Auto-tears __FilterToConsumerBinding bindings from anomalies. Hardens local WinRM listeners.
LSASS Credential Access	James Chen 58% (Active)	91	Windows Core Engine: Dynamically enforces Kernel Process Guard via RunAsPPL=1. Updates EDR block-rules to isolate modified procdump binaries immediately via hashes.
Registry Persistence Audit	Priya Sharma 100% (Done)	67	Continuous Baseline Loop: Enforces target desired state configuration over Run keys/AppInit DLLs. Auto-reverts deviations without human approval.
DNS Exfiltration Detection	Marcus Thompson 45% (Active)	78	Multi-Cloud Edge Engine: Syncs with Azure Firewall / AWS Route 53 resolvers. Auto-injects synthetic black-hole routing rules for bursts of TXT queries to untrusted TLDs.
Scheduled Task Implant	Elena Volkov 30% (Paused)	55	Cross-OS Cron Engine: Audits obfuscated arguments across Win Scheduled Tasks & Linux crontabs. Auto-suspends tasks matching high-entropy command parameters.
Kerberoasting Campaign	David Okonkwo 5% (Planning)	88	Hybrid Identity Engine: Automatically rotates weak SPN account keys within Managed Service Accounts framework. Auto-deploys honey-pot SPN accounts to trap attackers.
PowerShell Remote Hunt	Marta Kowalski 100% (Done)	74	Host Hardening Module: Forces block rules on unauthorized WinRM cross-subnet sessions. Dynamically auto-re-enables script block logging via GPO/Registry enforcement.
Cloud Metadata Service Abuse	Sarah Kim 40% (Cancelled)	--	Multi-Cloud Guard: Enforces IMDSv2 mandatory compliance on AWS/Azure instances. Dynamically strips wildcard IAM policies from instances violating network boundaries.

2. Unified Multi-SIEM / EDR Orchestration Abstract Layer

To support fully autonomous orchestration across heterogeneous vendor environments, the agent framework handles security event streaming and API abstraction natively. It wraps crowdstrike, defender, splunk, and sentinel commands into a zero-trust telemetry handler.

```
# Unified Autonomous Response Blueprint Schema Abstract
class CrossPlatformRemediationEngine:
    def __init__(self, target_host, os_flavor, provider_context):
        self.host = target_host
        self.os = os_flavor.lower() # windows, linux, darwin
        self.cloud = provider_context # aws, azure, gcp, hybrid

    def isolate_network_node(self):
        if self.cloud in ['aws', 'azure', 'gcp']:
            self.trigger_cloud_security_group_quarantine()
        if self.os == 'windows':
            self.execute_edr_isolate_api("Defender/CrowdStrike")
        elif self.os in ['linux', 'darwin']:
            self.execute_native_iptables_nftables_isolation()

    def kill_malicious_persistence(self, vector_payload):
        if self.os == "windows":
            # Direct WMI Object purge over root\subscription
            return "Get-CimInstance -Namespace root\subscription -ClassName __EventFilter | Remove-CimInstance"
        elif self.os == "linux":
            # Crontab verification loop
            return "sed -i '/' + vector_payload + '/d' /etc/crontab"
```

3. Deep-Dive Cross-OS Multi-Cloud Active Mitigations

A. WMI Lateral Movement Active Remediation (Windows Ecosystem)

The machine-agent establishes a real-time monitor via local RPC over WinRM to auto-purge subscription models without requiring administrative interaction. When a consumer binding matches no verified corporate baseline, the binding is cleanly severed.

```
# Autonomous WMI Event Consumer Disconnect
$BadFilter = Get-CimInstance -Namespace root\subscription -ClassName __EventFilter -Filter "Name like '%Malicious%'"
$Bindings = Get-CimInstance -Namespace root\subscription -ClassName __FilterToConsumerBinding
foreach ($B in $Bindings) {
    if ($B.Filter.Name -eq $BadFilter.Name) {
        Remove-CimInstance -InputObject $B -Verbose
        Remove-CimInstance -InputObject $BadFilter -Verbose
    }
}
```

B. Cloud Identity Metadata Service Protection (AWS / Azure / GCP Hybrid)

To counter token harvest scenarios, the engine updates cloud-native metadata access. For instance, instances are forced into IMDSv2 configuration with an automated token hop count limit of 1.

```
# AWS CLI / SDK Platform Automation Implementation
aws ec2 modify-instance-metadata-options \
  --instance-id i-00123456789abcdef \
  --http-tokens required \
  --http-put-response-hop-limit 1 \
  --endpoint enabled
```

C. Cross-Platform Linux / macOS Persistence Neutralization

For multi-OS coverage, the autonomous worker crawls all cron contexts, systemd services, and macOS launch daemons. High-entropy structures, obfuscated scripts, and binary drop paths are auto-quarantined.

```
# Systemd Malicious Service Purge Pipeline
systemctl stop compromised_agent.service
systemctl disable compromised_agent.service
rm -f /etc/systemd/system/compromised_agent.service
systemctl daemon-reload
```

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